Fraunhofer Institute for Open Communication Systems

DIN

DIN NIA-34 Working Group ODF / Open XML Translation

Agenda

1. Goals of the DIN NIA-34 Working Goup
2. Document formats – General Requirements
3. Why " translation" and not only " mapping"
4. Helping Interoperabilty:
   1. Identifying the differences between the formats –
   2. Translation criteria and examples

Goal of DIN NIA-34 Working Group

- Founded by DIN (Deutsches Institut für Normung) in May 2007
- Convenor - Gerd Schürmann (Fraunhofer FOKUS)
- Membership - the Working Group is open and is inviting international experts who would like to contribute to the Working Group's findings.
- Language - the Working Group conducts its work in English
- Current Members: PC-Ware Information Technologies AG, Microsoft Deutschland GmbH, OPENLiMiT Holding AG, Dialogika GmbH, Ecma TC45 liaison, Novell , Independent experts
- Main goal:
  - Identify in detail the differences between ODF and Open XML that can help Interoperability
  - Development of a technical report (DIN) concerning translation of ODF /Open XML in both directions.
- Methodology:
  - The translatability of tags and structures will be shown using use cases that describe certain scenarios and the round trip behavior.
  - In particular, best practices will be created for the areas where the document model use different functionalities or patterns

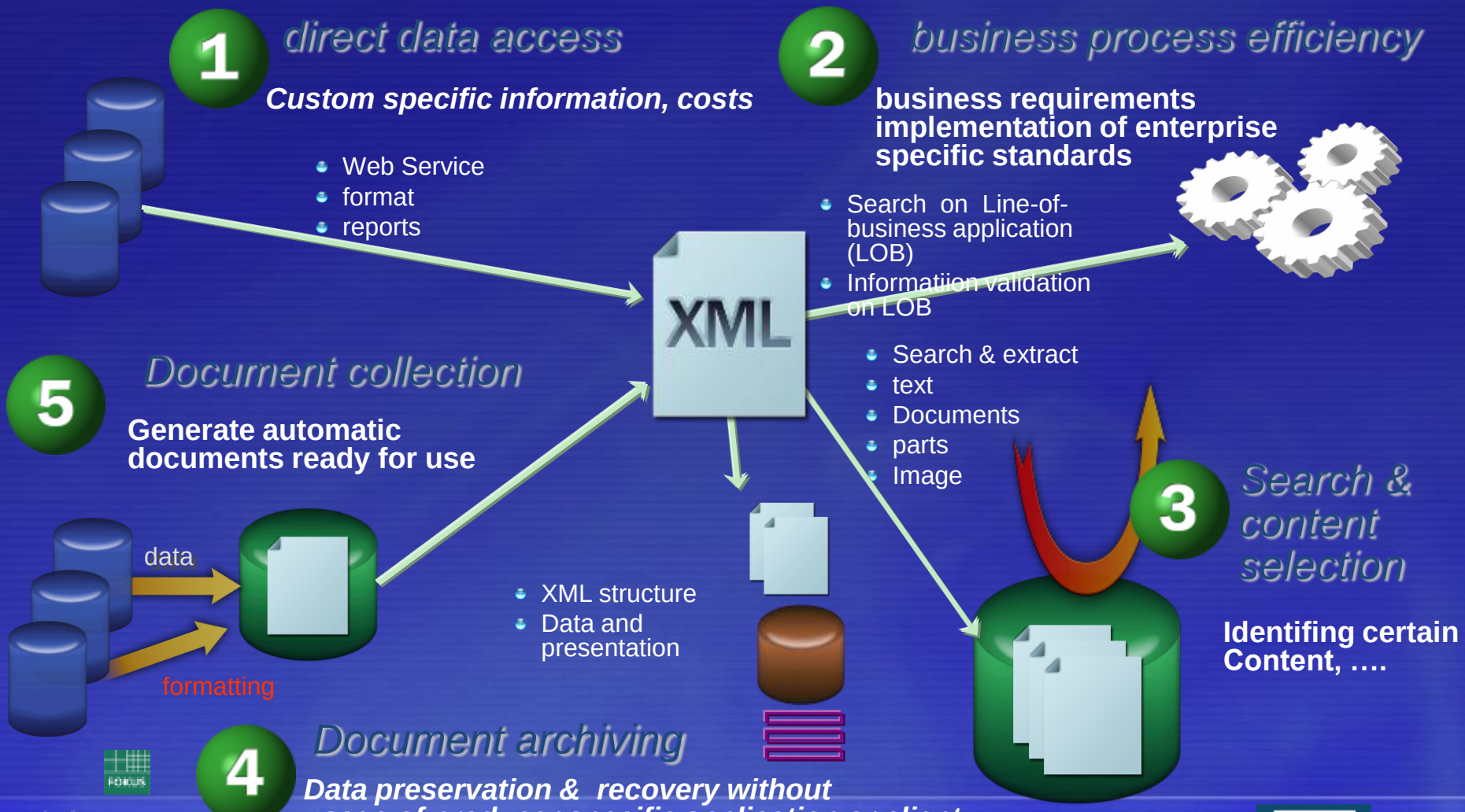
Program of work of DIN NIA-34 WG

- Working Group created in May 2007
- Conference calls and Face 2 Face meetings
- Timeframe
  - February 2008 – First Draft of Technical Report
  - July 2008 – Second Draft of Technical Report
  - February 2009 – Final Draft of Technical Report
- How to participate
  - Contact the WG Convenor - Gerd Schürmann (Fraunhofer FOKUS) at Gerd.Schuermann@fokus.fraunhofer.de
- Information on the work of the Working Group
  - Public Announcement: http://www.fokus.fraunhofer.de/fokus/fokus/presse/meldungen_fokus/2007/05/DIN-E.pdf
  - First Draft of Technical Report: http://www.nia.din.de/cmd?level=tpl-artikel&subcommitteeid=94340957&contextid=nia&cmstextid=74610&languageid=en

Requirements for Document Formats

- ODF / Open XML
  - Both document formats have to satisfy different requirements
  - Both document formats are open standards
  - Custom-made requirements concern the exchange of documents. There is a way of exchanging documents , the criteria for this are graphical fidelity , layout and structure.

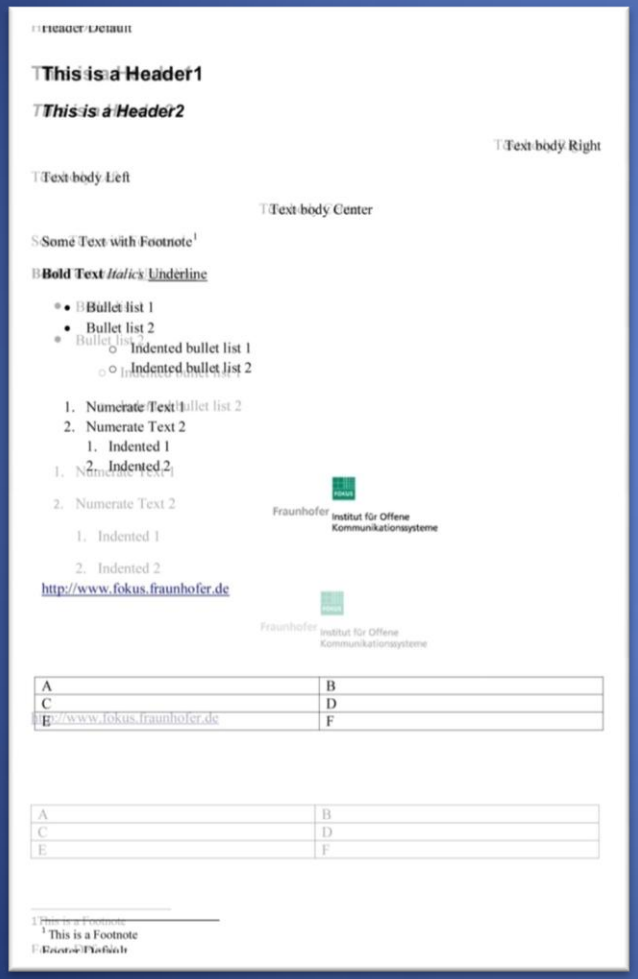
Document Formats
- the document landscape



Document Formats

- why “ translation“ and not only “ mapping“?-

Word-prozessing ML → ODF 1.0



Header Default

This is a Header1

This is a Header2

Text body Right

Text body Left

Text body Center

Some Text with Footnote[1]

Bold Text *Italics* Underline

- Bullet list 1
- Bullet list 2
  - Indented bullet list 1
  - Indented bullet list 2

1. Numerate Text 1
2. Numerate Text 2
   1. Indented 1
   2. Indented 2

http://www.fokus.fraunhofer.de

A	B
C	D
E	F

[1] This is a Footnote

ODF 1.0→ Word-processing ML

Header Default

This is a Header1

This is a Header2

Text Body Right

Text body Left

Text body Center

Some Text with Footnote[1]

Bold Text *Italics* Underline

- Bullet list 1
- Bullet list 2
  - Indented bullet list 1
  - Indented **bullet** list 2

1. Numerate Text 1
2. Numerate Text 2
   1. Indented 1
   2. Intended 2

http://www.fokus.fraunhofer.de

A	B
C	D
E	F

[1] This is a Footnote

Footer Default

Identifying the differences between the formats - Criterias

Conversion Type

One-time conversion

- ODF -> OOXML
- OOXML -> ODF

Round-trip conversion

- ODF -> OOXML -> ODF
- OOXML -> ODF -> OOXML

Functionality

- Graphical fidelity
- Editability / Revisability, Structure and layout
- Document Automation
- Preservation of meta-information

Use Cases

- Use Cases references to daily business needs
- Use Cases define the limitation of the report

Use Case (1) …

The Role of Leadership, by John Marketer

In an organization where there is faith in the abilities of formal leaders, employees will look towards the leaders for a number of things. During drastic change times, employees will expect effective and sensible planning, confident and effective decisions, and regular, complete communication that is timely. Also during these times of change, employees will perceive leadership as supportive, concerned and committed to their welfare, while at the same time recognizing that tough decisions need to be made. The best way to summarize is that there is a climate of trust between leader and the rest of the team. The existence of this trust brings hope for better times in the future, and that makes coping with drastic change much easier.

In organizations characterized by poor leadership, employees expect nothing positive. In a climate of distrust, employees learn that leaders will act in indecipherable ways and in ways that do not seem to be in anyone's best interests. Poor leadership means an absence of hope, which, if allowed to go on for too long, results in an organization becoming completely non-functioning. The organization must deal with the practical impact of unpleasant change, but more importantly, must labour under the weight of employees who have given up, have no faith in the system or in the ability of leaders to turn the organization around.

Leadership before, during and after change implementation is the key to getting through the swamp. Unfortunately, if haven't established a track record of effective leadership, by the time you have to deal with difficult changes, it may be too late.

Gelöscht: JOHN MARKETER

Gelöscht: -making

Kommentar [e1]: Please explain more in detail!

Kommentar [e2]: Need for concrete examples.

Gelöscht: THE

Use Case (2) …

Scenario description

After the co-author had sent back the revised version of the article John can start to adapt his text regarding comments and proposed changes.

The comments are simple text notes and have to be applied manually. The tracked changes could be applied automatically for the whole document if John agrees with the revision.

This kind of application supported workflow highly alleviates the revision of documents with multiple authors.

The foundation for this document lifecycle is the proper conversion of metadata from one standard into the other to keep comments and proposed changes

The rating …

The tables in the following subsections as well as those in section 3 summarize the availability of various functionalities for each of the two document formats as well as an estimate of the functionalities' "translatability", defined as follows:

Low: Either one of the formats does not contain this functionality at all, or the way the functionality is realized differs a lot, so that conversion of this feature is impossible without information loss.

Medium: Functionalities categorized as having a translatability of "Medium" usually are supported in both formats, although some aspects might differ and workarounds may be required. The "Notes" column provides more details.

High: These functionalities are supported by both formats and should cause no or very few problems during conversion.

Functionalities

- Explicit references to use cases
- Mostly in table form
  - Description of the (sub-) functionality
  - Which standard(s) implement it?
  - Translatability rating
  - Notes

Functionalities - examples

Functionality	Sub-Feature (if applicable)	Office Open XML	Open Document	Trans-latability	Notes
Bold text (font weight)		Yes	Yes	Medium	In addition to bold, OpenDocument allows font weight to be specified numerically (100-900). However e.g. OpenOffice.org only supports bold/non-bold.
Text color					
	RGB	Yes	Yes	High	
Font selection					
	By font name	Yes	Yes	High	
	By font family	Yes	Yes	High	
	Theme fonts	Yes	No	Low	OpenDocument does not support the concept of document themes
Italic text		Yes	Yes	Medium	OpenDocument supports italic and oblique text, Office Open XML makes no such distinction
Kerning		Yes	Yes	High	
Text language		Yes	Yes	High	
Raised / lowered text		Yes	Yes	Medium	Office Open XML uses absolute values, OpenDocument uses percentages. This may lead to conversion problems.
Underline		Yes	Yes	Medium	Office Open XML allows single and double underline. OpenDocument offers more fine-grained control of underline options and styles.

Functionalities - examples (2)

- Rating often not immediately obvious
- Implementations diverge from specs
  - TR focuses on specifications
  - Offers some hints/workarounds where appropriate

Functionality	Office Open XML	Open Document	Trans-latability	Notes
Bold text (font weight)	Yes	Yes	Medium	In addition to bold, OpenDocument allows font weight to be specified numerically (100-900). However e.g. OpenOffice.org only supports bold/non-bold.
Theme fonts	Yes	No	Low	OpenDocument does not support the concept of document themes

Functionalities - examples (3)...

Functionality	Sub-Feature (if applicable)	Office Open XML	Open Document	Translatability	Notes
Text insertion		Yes	Yes	High	
Text deletion		Yes	Yes	High	
Formatting changes		Yes	Yes	Medium	OpenDocument only records the fact that a change has occurred. However, no further information is recorded, so that it is impossible to reconstruct the previous state.
Comments		Yes	No	Medium	Office Open XML allows adding comments to arbitrary text ranges. This is not supported by OpenDocument, however similar functionality may be provided by inserting Notes (which are associated with a point in the text, not a range).
Text highlighting		Yes	No	High	Although OpenDocument does not support text highlighting, the functionality may be emulated by setting the text background color (see the section on text formatting)
Meta Data					
	Name	Yes	Yes	High	
	Date / Time	Yes	Yes	High	
	Author shorthand for comments	Yes	Yes	High	

What the TR will provide and status

- Architecture of the container formats
- Complete coverage of selected functional aspects

- Section on text processing essentially complete
- Use cases for spreadsheets and presentations drafted

Summary

- Next steps:
  - Spreadsheets
  - Presentations
- Formats differ even in seemingly obvious aspects
- Feedback and suggestions are welcome!

Actions

- We as the SC34 Mirror in Germany call for support of other NB's that are interested on that topic
  - Request to SC34 in Oslo!
- Feedback and suggestions are welcome!
- Please contact:

Secretariat ISO/IEC JTC 1/SC34
DIN German Institute for Standardization
DE-10772 Berlin
Tel. + 49 30 2601 2652
Fax + 49 30 2601 1723
E-mail Jan.Dittberner@din.de

Thank you for your attention!